

HPS-GPR v5.5.0

Beam energy and the shared mass structures

11 September 2026 | Diagnostic extension of v5.0.4

What the comparison shows

The different beam energies offer two distinct questions: whether structures move in mass, and whether an additional energy-dependent response could explain their different fitted rates. The simple $m \propto \sqrt{E}$ and $m \propto E$ mappings do not preserve a positive three-campaign feature associated with the 2016 90–92 MeV excess. At fixed 92 MeV all three fitted amplitudes are positive, but their magnitudes disagree with the inherited common-coupling model.

An empirical rate correction at fixed 92 MeV can absorb most of that disagreement. It requires a steep additional response, approximately $E^{-2.90}$, after the existing normalization. This describes the three selected amplitudes; it does not identify a production mechanism, establish coupling running, or validate a particle interpretation.

Inputs and beam energies

Dataset	Beam energy [GeV]	W target [% X_0]	Fitted mass range [MeV]
2015 full	1.056	0.125	19–100
2016 full	2.30	0.125	39–180
2021 10%	3.74	0.625	50–250

Energies and target thicknesses come from the HPS run table [2]; the original 2015 publication independently gives 1.056 GeV [1]. The fitted ranges above belong to this v5.0.4 analysis, not the published searches. The tracker and trigger also changed before 2021. These campaigns therefore do not constitute a controlled energy scan.

A reproducible mass comparison

For each dataset i , define

$$m_i = m_{\text{ref}}(E_i/2.30 \text{ GeV})^\alpha, \quad \alpha \in \{0, \tfrac{1}{2}, 1\}.$$

The reference is 2016. Anchors 66, 72, 76 and 92 MeV retain the parent’s comparisons; 90, 102 and 117 MeV cover the leading 2016 excess, deficit and next separated excess. All are retrospective choices. The half-MeV saved scans are replotted without altering their fitted values. At the anchors, 47 distinct deterministic standalone fits evaluate the exact mapped masses with the inherited moving masks, resolutions, kernel interpolation and Poisson/Gaussian background constraints.

The 2015 extension above 90 MeV retains its frozen 90 MeV kernel and 14–135 MeV training support. Unsupported masses remain untested. No new data, toys, exponent search over mass mappings, or discovery calibration is introduced. The signed root $r = \text{sgn}(\hat{A})\sqrt{2(\text{NLL}_0 - \text{NLL}_{\text{free}})}$ describes local fit response, including deficits; it is not a calibrated global significance.

The common-coupling comparison with energy-scaled masses

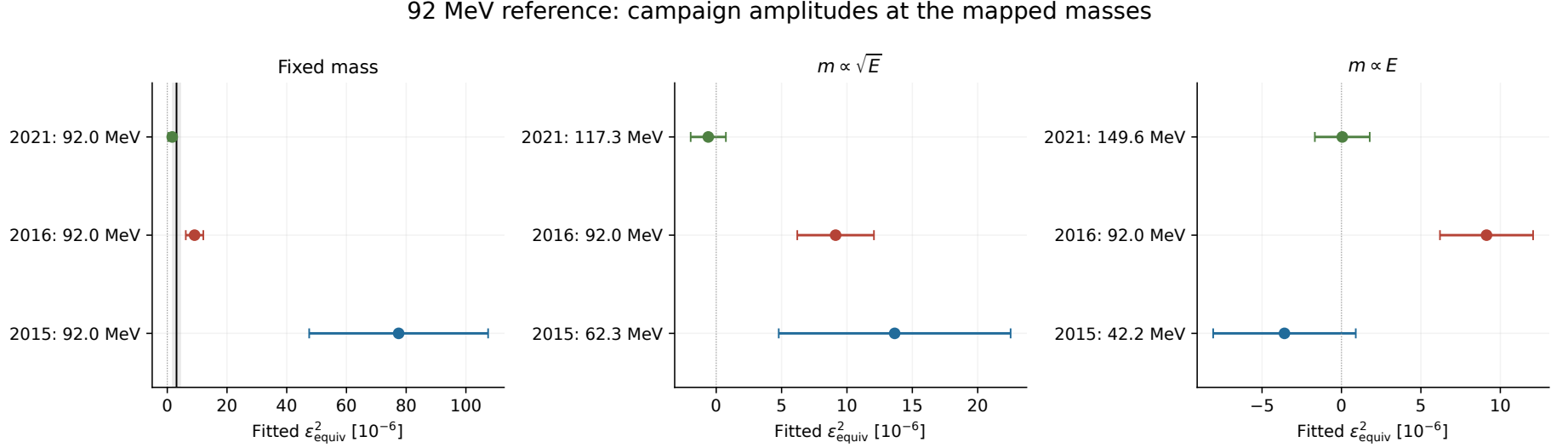


Figure 1: Extension of the parent’s campaign-amplitude layout at a 92 MeV reference. Bars are local curvature errors. In the fixed-mass panel, the black line and gray band reproduce the common fit, $(3.096 \pm 1.229) \times 10^{-6}$. For $\alpha > 0$, the points describe different physical masses and $\epsilon_{\text{equiv}}^2$ is only the inherited dark-photon-equivalent template normalization. No common particle coupling is fitted across those different masses.

α	Dataset	Mapped mass [MeV]	$\epsilon_{\text{equiv}}^2$ [10 ⁻⁶]	Signed r
0	2015	92.000	77.483 ± 29.958	+2.590
0	2016	92.000	9.132 ± 2.932	+3.115
0	2021	92.000	1.655 ± 1.354	+1.222
0.5	2015	62.338	13.649 ± 8.867	+1.540
0.5	2016	92.000	9.132 ± 2.932	+3.115
0.5	2021	117.317	-0.606 ± 1.347	-0.450
1	2015	42.240	-3.588 ± 4.488	-0.799
1	2016	92.000	9.132 ± 2.932	+3.115
1	2021	149.600	0.052 ± 1.727	+0.030

At fixed 92 MeV, the common-coupling fit has a twice-NLL penalty of 11.552 relative to separate amplitudes (two additional parameters). Square-root scaling instead samples a small 2021 deficit; linear scaling samples a 2015 deficit and almost no 2021 response. These comparisons do not provide a calibrated preference between mass-scaling hypotheses: their local windows and background constraints differ. At exactly 90 MeV, even the fixed-mass 2021 fit is negative ($r = -0.420$); “shared near 90 MeV” describes a region, not three coincident fitted maxima.

Could beam energy explain the rate mismatch at fixed mass?

Hold all masses at 92 MeV and introduce an additional relative response,

$$\lambda_i = b_i + L_i\theta_i + \mu K_i(92)R_i, \quad R_i = (E_i/2.30 \text{ GeV})^\beta.$$

Here K_i is the inherited signal response in counts per unit ϵ^2 , including its prompt-density conversion and effective radiative fraction. Thus R_i represents missing response *after* that conversion. A theoretical production power must not be added again without accounting for what K_i already includes [3, 4].

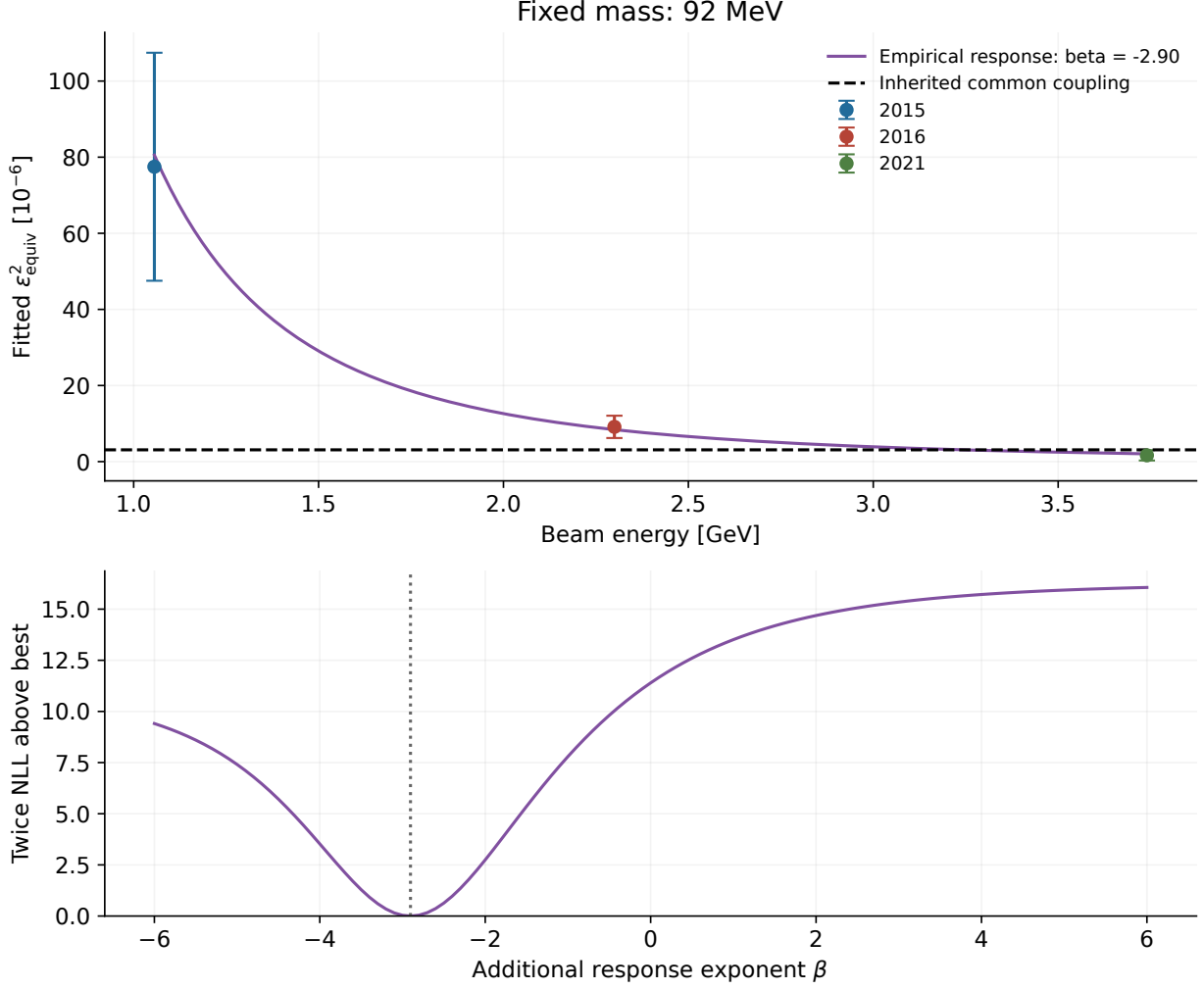


Figure 2: Empirical residual-response fit on the same fixed 92 MeV data and likelihood windows. The lower panel profiles the common nonnegative amplitude for each β ; its vertical values are objective differences, with no confidence or significance thresholds assigned.

The fitted slope is $\beta = -2.904$, with reference amplitude $\mu = 8.403 \times 10^{-6}$ and response factors (9.585, 1, 0.244) for 2015, 2016 and 2021. It reduces twice NLL by 11.396 relative to $\beta = 0$, leaving 0.157 relative to three separate amplitudes. The search range was $[-6, 6]$; neither the slope bound nor zero amplitude was reached. Two parameters describing three selected rates is close to a saturated comparison. The large required response factors demand independent detector or normalization support before a physical use.

Other excesses and deficits

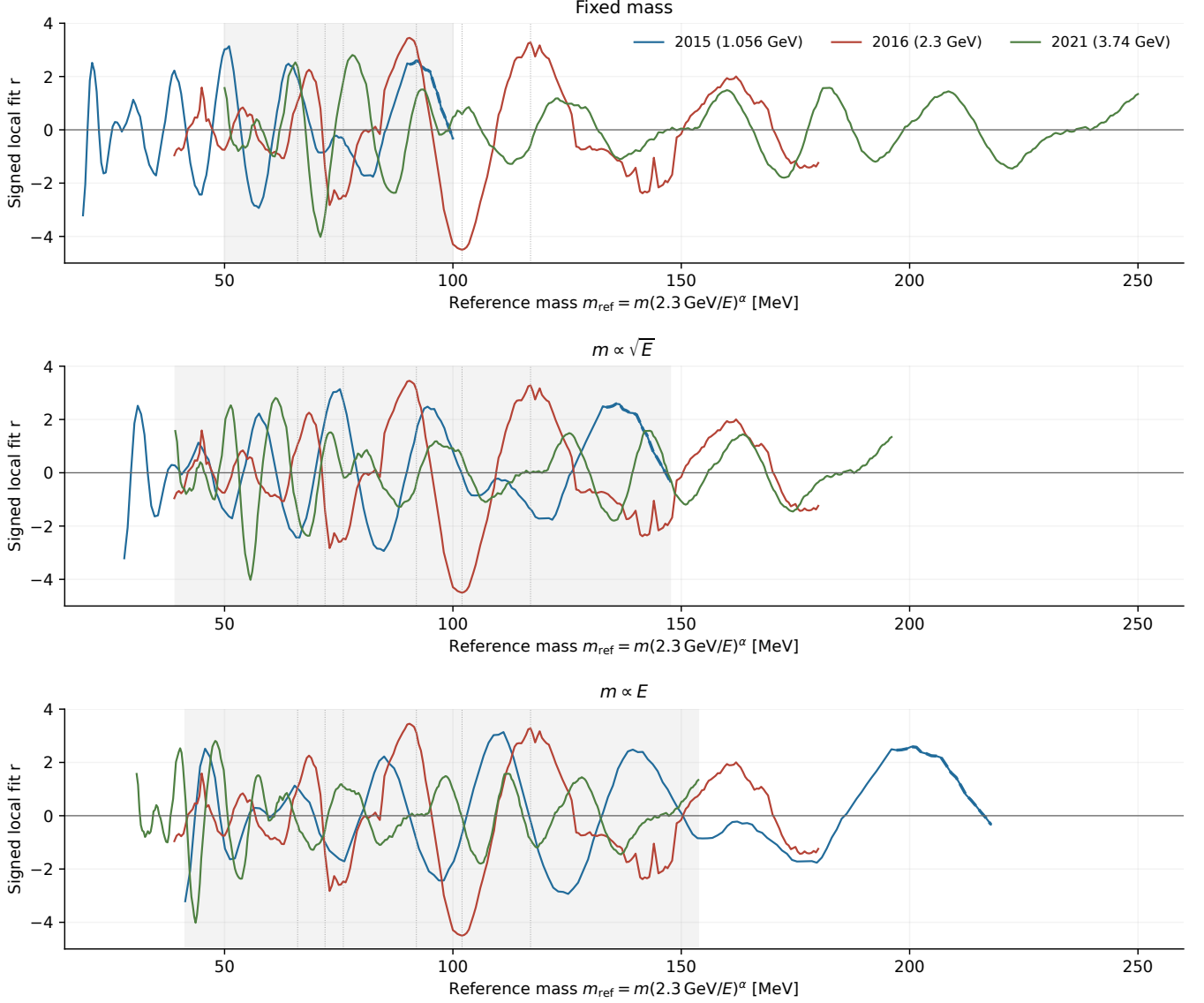


Figure 3: The same saved signed scans on three reference-mass coordinates. Gray shading marks support common to all three campaigns: 50–100, 39.21–147.58 and 41.38–153.74 MeV, respectively. The dashed part of the blue trace is the inherited 2015 extension above native 90 MeV. Lines join saved half-MeV samples; anchor fits elsewhere in this note are evaluated at their exact mapped masses.

The fixed 66 MeV comparison is positive in all campaigns; square-root scaling changes both non-reference campaigns to deficits. The fixed 72 MeV comparison is negative in all campaigns; square-root scaling changes both non-reference campaigns to excesses. Linear scaling retains negative signs at that anchor, but the 2021 response is weak. At 76 MeV the campaigns already have opposing signs. At the 2016 117 MeV excess, neither alternative mapping produces two additional positive responses. Some visual coincidences occur elsewhere; the overlays do not establish a universal scaling law. An auxiliary catalog records all 55 sign-consistent interior extrema with $|r| \geq 1$ without fitting a matching rule.

Physical interpretations and their limits

A fixed-mass boson. A kinetically mixed dark photon is a concrete model for a narrow e^+e^- resonance [4]. Its pole mass is independent of the beam energy. Production, boost, angular acceptance and decay acceptance change with energy. This invariant-mass statement also applies to other bosons, whose production and angular responses would require their own calculations. A shared mass alone does not explain the fitted rates or distinguish a resonance from a reconstruction effect.

Kinematic or acceptance structure. The relativistic pair-mass relation used by HPS is [3]

$$m_{ee} = 2\sqrt{p_+p_-} \sin(\theta_{+-}/2) \simeq E_{\text{beam}}\sqrt{x_+x_-} \theta_{+-}, \quad x_{\pm} = p_{\pm}/E_{\text{beam}}.$$

The small-angle limit on the right is derived here. Fixing the two momentum fractions and opening angle gives linear scaling. Fixing one absolute momentum, the other momentum fraction and the angle gives square-root scaling. These are conditional kinematic limits, not detector simulations. Magnetic settings, geometry, trigger thresholds and selection changes can invalidate either assumption. QED tridents, converted bremsstrahlung and mispaired tracks motivate such controls; these sources do not by themselves predict a universal narrow 90 MeV pole.

Secondary-positron annihilation. Resonant annihilation on a free electron at rest obeys [5]

$$s = 2m_e(E_+ + m_e), \quad E_+^{\text{res}} = m_X^2/(2m_e) - m_e.$$

Using $m_e = 0.51099895$ MeV and $E_+ \leq E_{\text{beam}}$ gives nominal mass endpoints 32.860, 48.488 and 61.829 MeV. A 90 MeV state would require a 7.925 GeV positron in this approximation. This rules out the simple stationary-electron mechanism at 90 MeV for these beams, not every atomic-electron mechanism. High- Z electron momenta can extend the nominal mass reach [6]. Establishing an HPS contribution requires a tungsten electron momentum model, secondary-positron transport, material geometry and accepted decay calculation. Even in that production channel, a real state's mass stays fixed; the accessible range and rate vary.

What would discriminate the possibilities. Use event-level splits in opening angle, momentum fractions, absolute momenta, detector half, hit category and run period, with matched simulated selections. Check mass and momentum calibration with Moller pairs and full-energy electrons [3]. A reproducible mass shift with such controls would test a specific acceptance mechanism. A residual-response model would need to predict the factors in the fixed 92 MeV rate fit independently. The saved one-dimensional spectra cannot perform those tests or separate beam energy from the other campaign changes.

Sources and reproducibility

This study was developed with an independent physics-agent literature consultation and a separate numerical review. The primary literature supports the energies and model mechanisms; all HPS-GPR fit numbers in this note are derived from the pinned local v5.0.4 package, not from published excess claims.

The study copies its required input spectra and solver modules into `inputs/` and `engine/`. The input ledger records SHA-256 identities and the checkout commit. Run `scripts/run_study.py`, `scripts/rate_response.py`, and `scripts/report_tables.py`; compile `source/main.tex` with Tectonic. Figure PDFs are vector outputs and their numerical arrays are retained in `derived/`. The final package includes a checksum manifest and numerical and rendered-page QA.

All 47 standalone fits passed the inherited convergence and positive-expectation checks. Exact common-mass replays reproduce the parent’s amplitude-consistency results. The added rate fit uses identical data and constraints across β , so its likelihood differences are meaningful descriptive comparisons within that family. No such comparison is made between different α windows. Curvature bars, selected extrema, fitted response slopes and apparent alignments have no post-selection probability or coverage interpretation. The retained 2016 background qualifications and 2015 extension limitations continue to apply.

The supplementary amplitude figure covers all four inherited anchors. The independent audit’s 37 resolution-separated extrema are explicitly a thinned catalog; the main script’s 55 unthinned extrema use a different, documented display rule. Interpolated numbers in the audit are preliminary cross-checks; the exact continuous fits in `derived/anchor_fits.csv` are authoritative for this note.

References

- [1] P. H. Adrian et al. (HPS Collaboration). Search for a dark photon in electroproduced e^+e^- pairs with the Heavy Photon Search experiment at JLab. *Phys. Rev. D* 98, 091101 (2018). [10.1103/PhysRevD.98.091101](#).
- [2] N. Baltzell et al. (HPS Collaboration). The Heavy Photon Search Experiment. *arXiv:2203.08324*. [2203.08324](#).
- [3] P. H. Adrian et al. (HPS Collaboration). Searching for prompt and long-lived dark photons in electroproduced e^+e^- pairs with the heavy photon search experiment at JLab. *Phys. Rev. D* 108, 012015 (2023). [2212.10629v3](#).
- [4] J. D. Bjorken, R. Essig, P. Schuster and N. Toro. New Fixed-Target Experiments to Search for Dark Gauge Forces. *Phys. Rev. D* 80, 075018 (2009). [0906.0580](#).
- [5] L. Marsicano et al.. Dark photon production through positron annihilation in beam-dump experiments. *Phys. Rev. D* 98, 015031 (2018). [1802.03794v2](#).
- [6] F. Arias-Aragon, L. Darne, G. Grilli di Cortona and E. Nardi. Production of dark sector particles via resonant positron annihilation on atomic electrons. *Phys. Rev. Lett.* 132, 261801 (2024), corrected *arXiv v2*. [2403.15387v2](#).

Mapped-window extraction at the 92 MeV reference

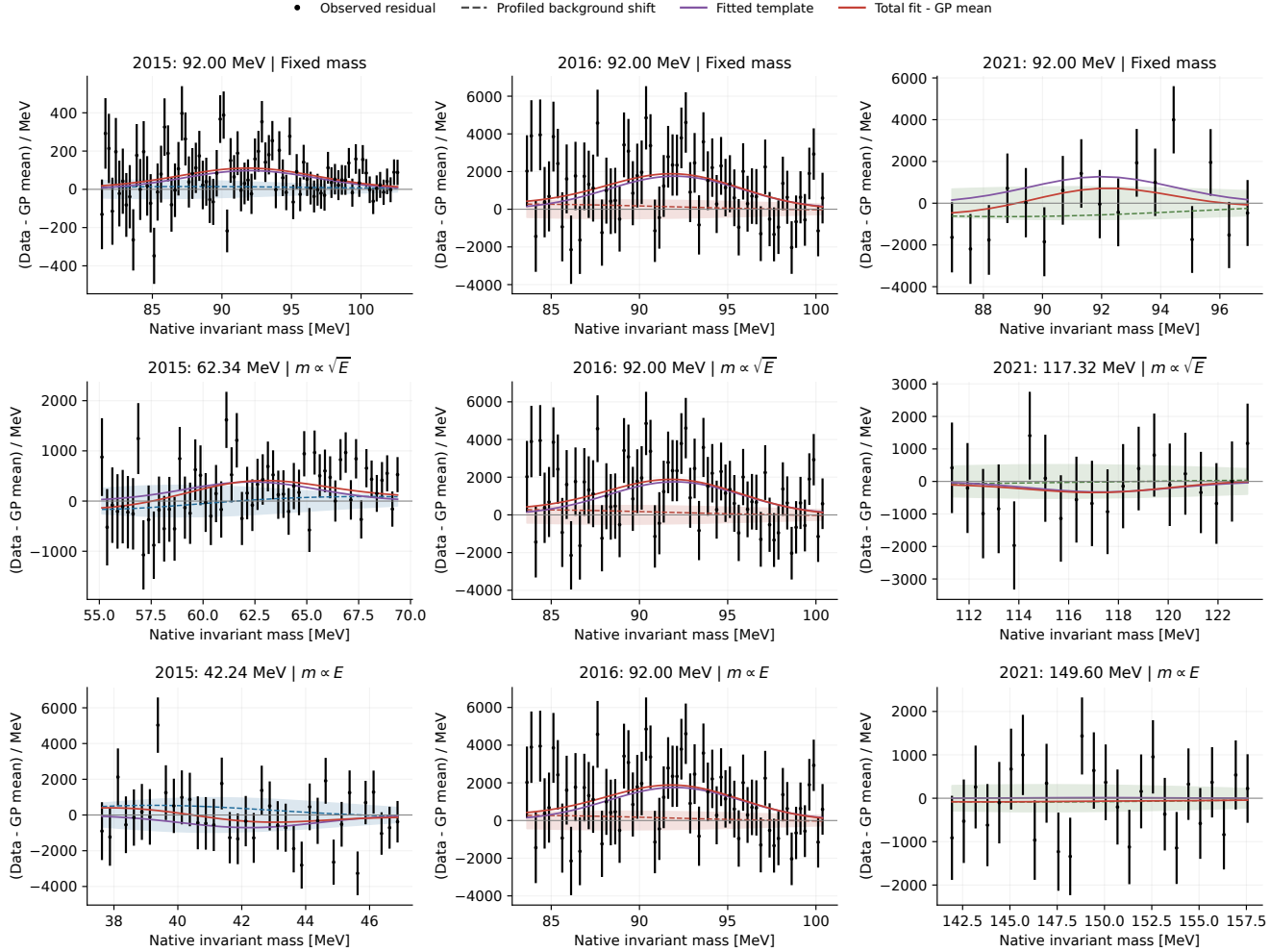


Figure 4: Native-bin residuals for the nine mapped fits, analogous to the parent’s common-coupling extraction display. Each panel here fits its own amplitude, including in the fixed-mass row. The data subtract the same unprofiled GP mean used for all curves in that panel. Bars show counting error; shading is the GP constraint width, not total residual or fitted-background uncertainty. Repeated 2016 panels are the same reference fit.

A common signed comparison for the other structures

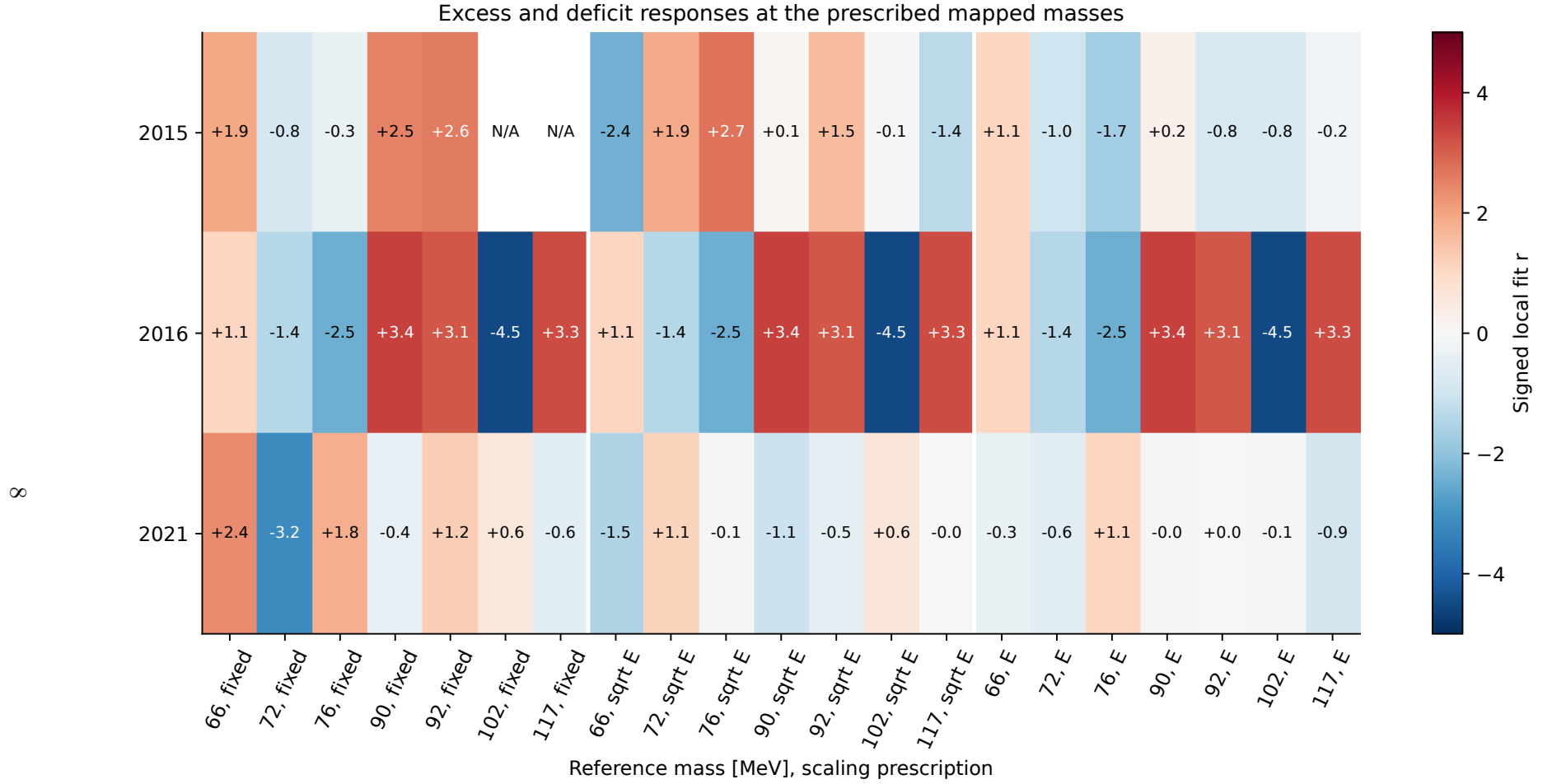


Figure 5: Direct fitted roots at the prescribed masses, retaining signs and unsupported cells. The columns do not represent independent tests; nearby mass fits are strongly correlated. No threshold in this display defines evidence for or against a physical model.